

Queues



❑ FIFO (First in First Out)

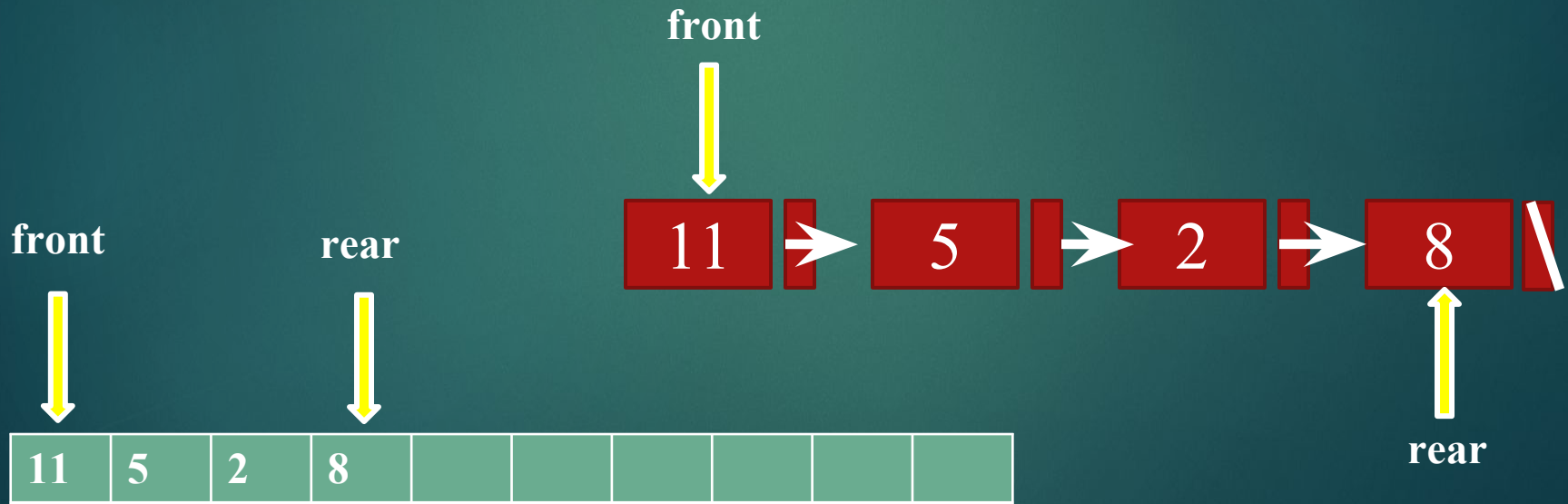
Inserted at one end and remove at another end. While in stack insert and remove on same side.

❑ Operation

- ▶ enqueue(x) insert at end
- ▶ dequeue (). Remove from front
- ▶ front(). Return the front element
- ▶ isEmpty(). Return true if empty else false

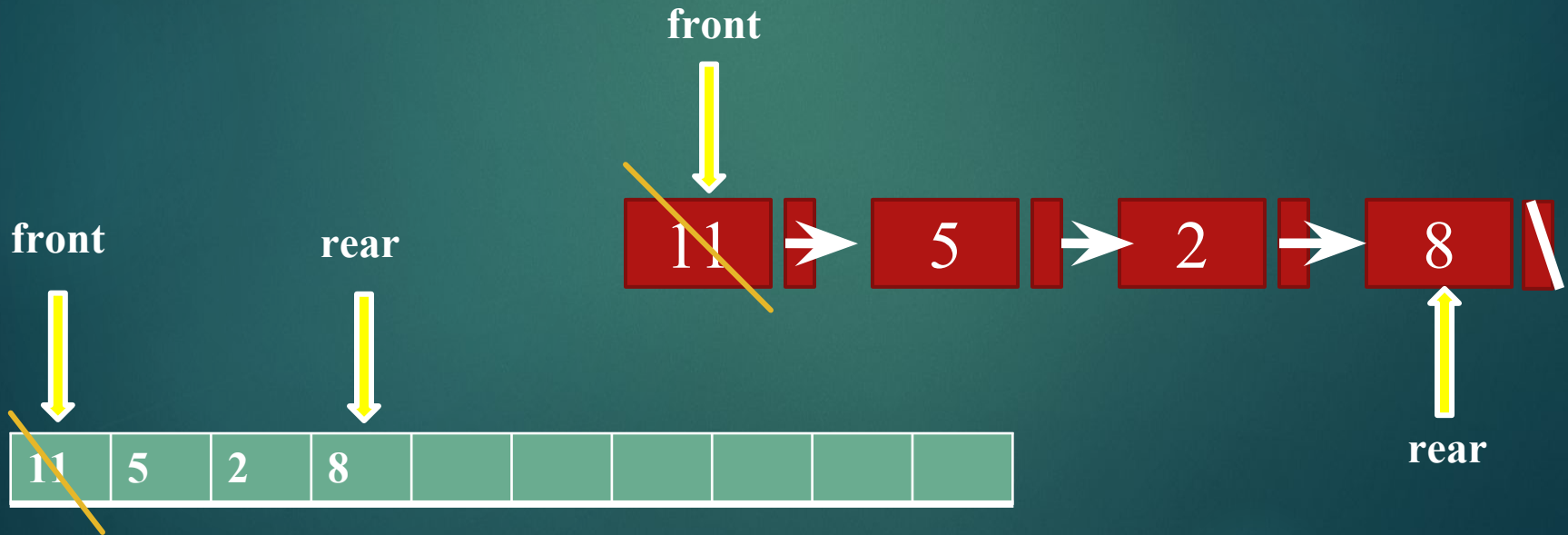
Implementing Queue

- ❑ Using linked List and Array:
- ❑ Insert and remove in constant time



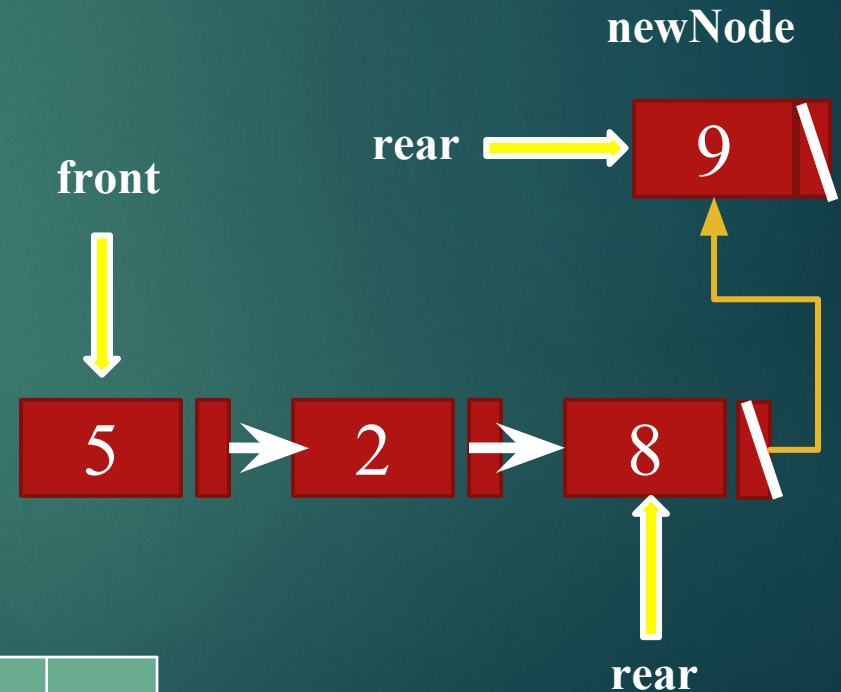
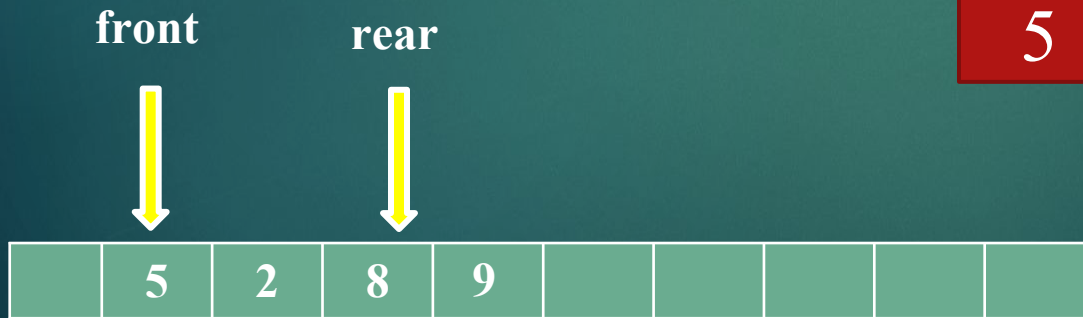
Implementing Queue

dequeue()

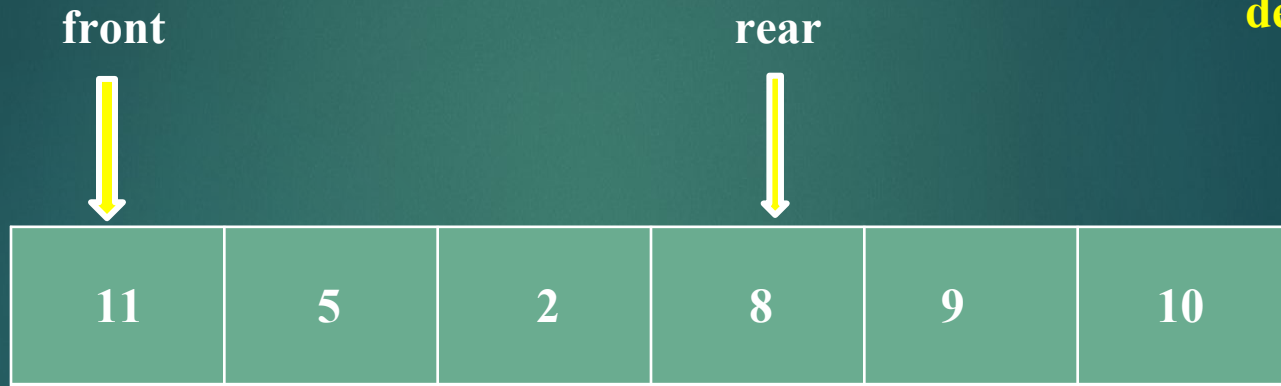


Implementing Queue

enqueue(9)



Implementing Queue using Array



dequeue()

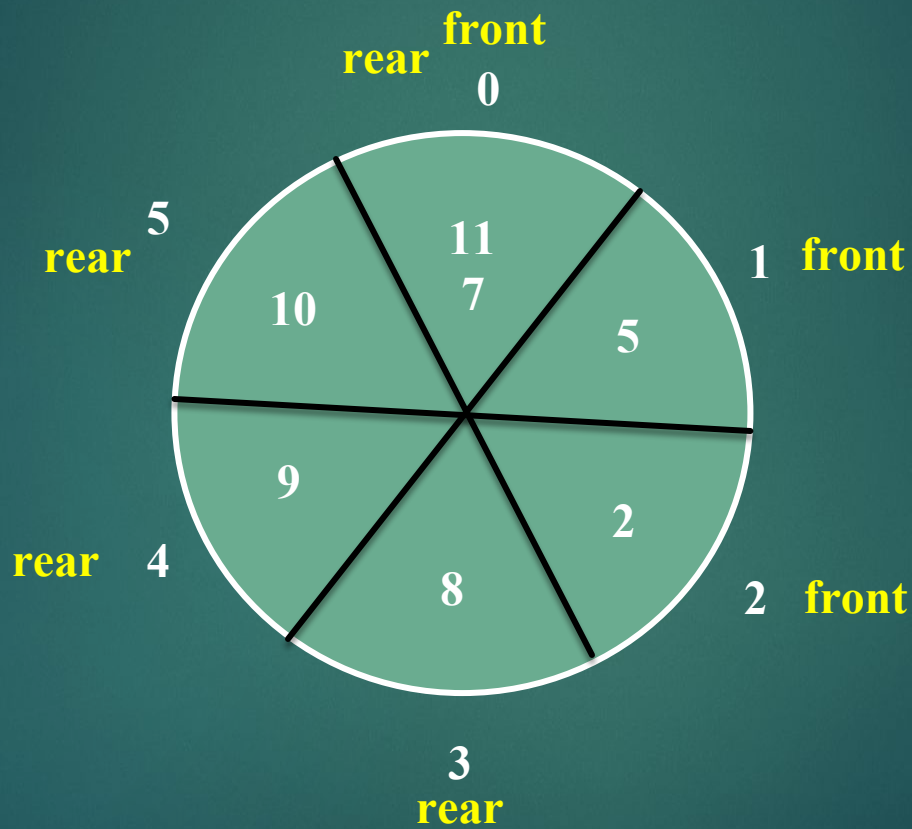
dequeue()

Capacity is Full

enqueue(9)

enqueue(10)

enqueue(7)

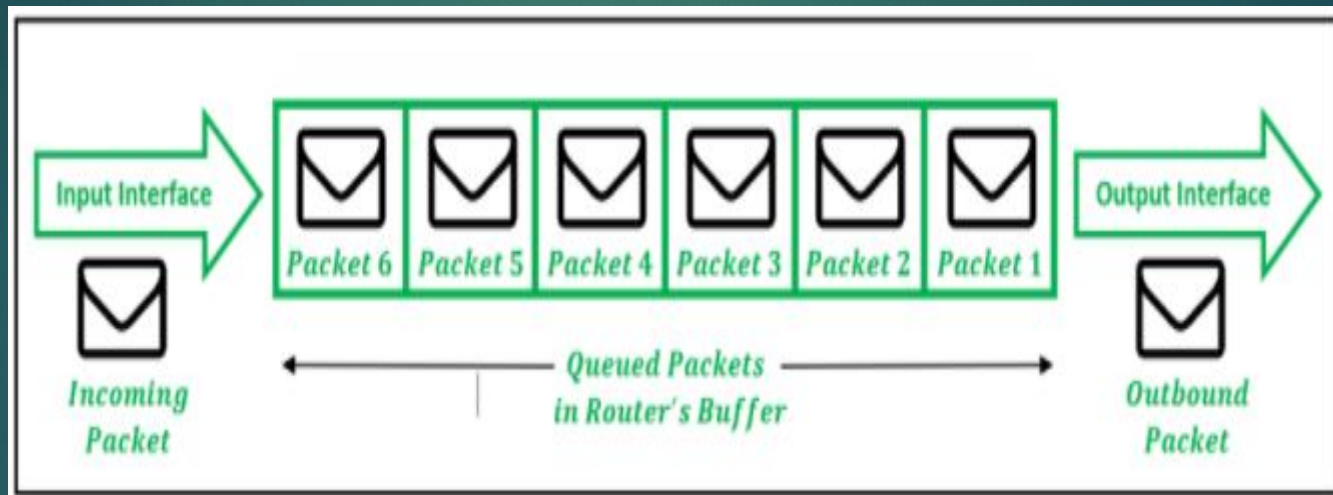


dequeue()
dequeue()
enqueue(9)
enqueue(10)
enqueue(7)

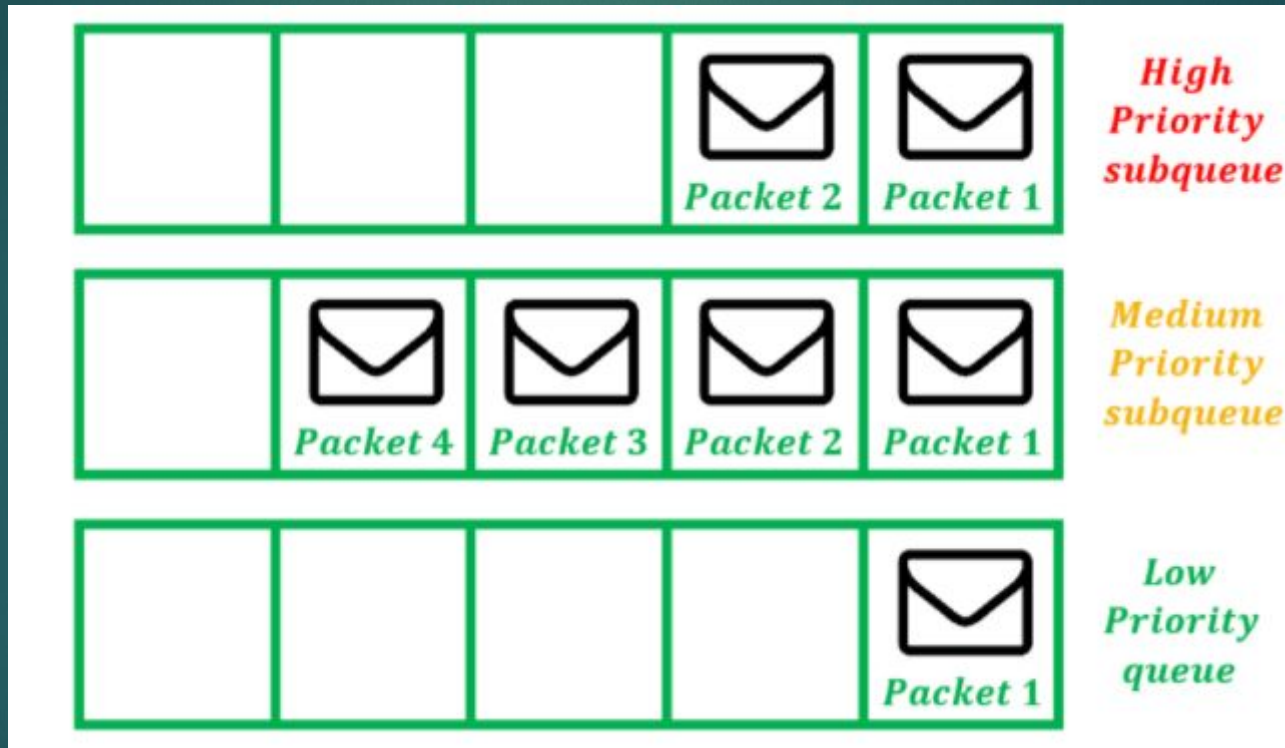


Queue Applications

Packet Queuing and Dropping in Routers

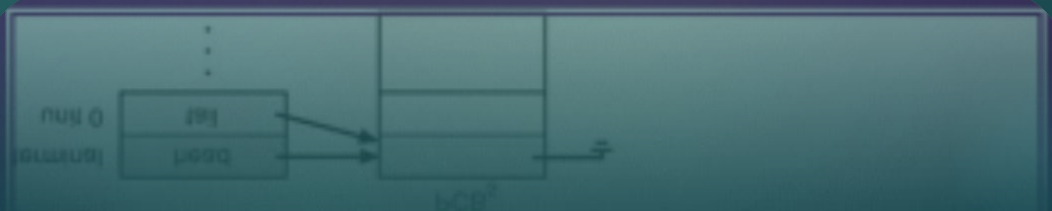
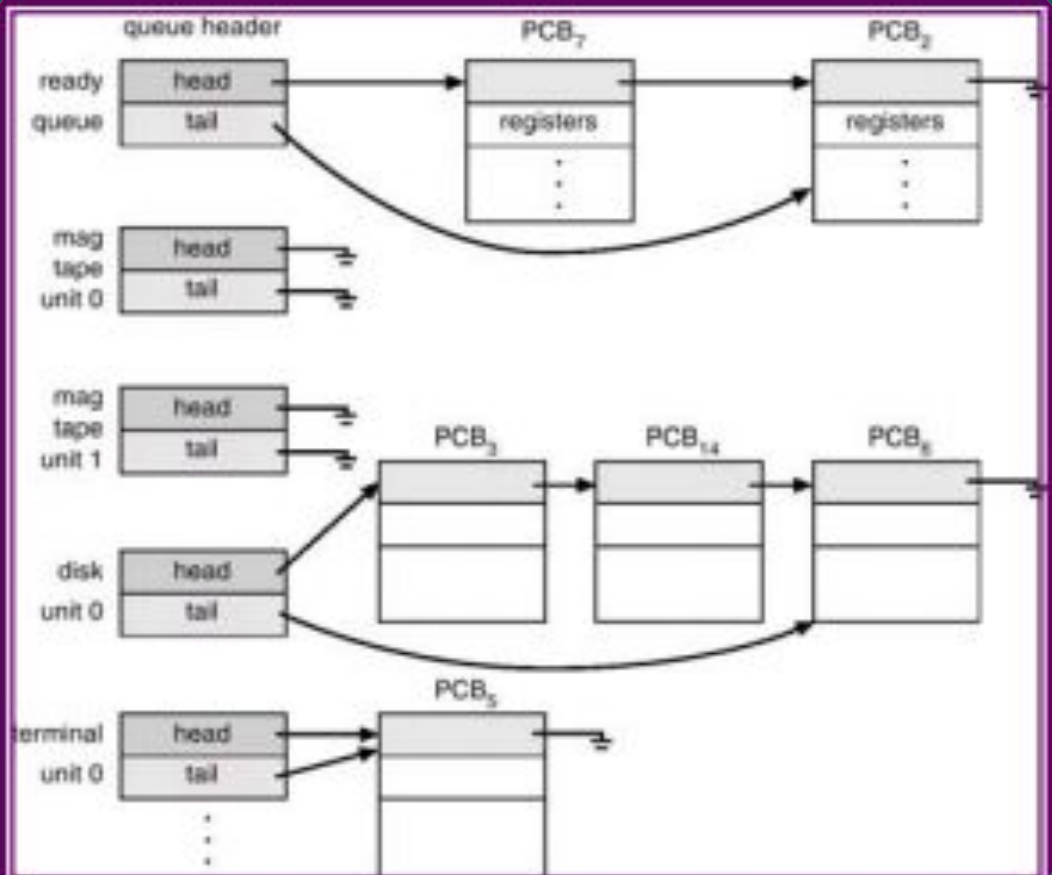


Packet Queuing and Dropping in Routers



Scheduling Queues:

- ❑ **Job Queue:** As processes enter the system, they are put into a job queue. This queue consists of all processes in the system.
- ❑ **Ready Queue:** The processes that are residing in main memory and are ready and waiting to execute are kept on a list called the ready queue. This queue is generally stored as a linked list. A ready-queue header contains pointers to the first and final PCBs in the list. Each PCB is extended to include a pointer field that points to the next PCB in the ready queue.
- ❑ **Device Queue:** When a process is allocated the CPU, it executes for a while, and eventually quits, is interrupted or waits for a particular event, such as completion of an I/O request. In the case of an I/O request, the device may be busy with the I/O request of some other process, hence the list of processes waiting for a particular I/O device is called a device queue. Each device has its own device queue.



CPU Scheduling:



- ▶ FIFO
- ▶ Shortest job first
- ▶ Shortest remaining job first
- ▶ Round Robin